

SOC L1 Readiness Test (Sections A, B, C & D)

Student Version – Answer key with explanations provided in the second half of the document.

Section A — Multiple Choice Questions (25)

1. *What is the primary role of a SOC L1 analyst?*

- A. Develop malware
- B. Monitor alerts and perform initial triage
- C. Write security policies
- D. Perform penetration testing

2. *Which malware encrypts files and demands payment?*

- A. Trojan
- B. Worm
- C. Ransomware
- D. Spyware

3. *Which attack uses one password across many accounts?*

- A. Brute Force
- B. Dictionary Attack
- C. Password Spraying
- D. Credential Stuffing

4. *What does hashing primarily provide?*

- A. Confidentiality
- B. Integrity
- C. Authentication
- D. Availability

5. *What does salting prevent?*

- A. Brute force attacks
- B. Rainbow table attacks
- C. Phishing
- D. DDoS

6. *Which tool is commonly used for credential dumping?*

- A. Wireshark
- B. Metasploit

- C. Mimikatz
- D. Burp Suite

7. Which Windows process stores authentication credentials?

- A. svchost.exe
- B. explorer.exe
- C. lsass.exe
- D. winlogon.exe

8. Which attack redirects users to a fake website without clicking a malicious link?

- A. Phishing
- B. Pharming
- C. Smishing
- D. Baiting

9. Moving between compromised systems inside a network is called?

- A. Privilege escalation
- B. Persistence
- C. Lateral movement
- D. Data exfiltration

10. Which encryption type uses two keys?

- A. Symmetric
- B. Asymmetric
- C. Hashing
- D. Encoding

11. Fraudulent emails to steal credentials represent which attack?

- A. Phishing
- B. Malware
- C. DDoS
- D. SQL Injection

12. What does IOC stand for?

- A. Indicator of Compromise
- B. Incident of Crime
- C. Internal Operations Control
- D. Internet Operations Center

13. Which attack uses leaked credentials from previous breaches?

- A. Brute force

- B. Credential stuffing
- C. Password spraying
- D. Dictionary

14. Which attack floods servers with traffic?

- A. Malware
- B. Phishing
- C. DDoS
- D. Brute Force

15. Which technique extracts credentials from system memory?

- A. Credential dumping
- B. Encryption
- C. Hash collision
- D. Tokenization

16. Targeted phishing aimed at a specific individual is called?

- A. Phishing
- B. Spear phishing
- C. Smishing
- D. Vishing

17. What does SIEM primarily do?

- A. Develop malware
- B. Log collection and analysis
- C. Replace antivirus
- D. Disable attacks

18. Which security principle ensures data is unchanged?

- A. Confidentiality
- B. Integrity
- C. Availability
- D. Authentication

19. Which protocol is used for remote Windows access?

- A. FTP
- B. RDP
- C. SMTP
- D. DNS

20. Which authentication uses multiple verification factors?

- A. Single factor
- B. MFA
- C. Password only
- D. Tokenization

21. What is alert triage?

- A. Ignoring alerts
- B. Investigating and prioritizing alerts
- C. Deleting logs
- D. Disabling SIEM

22. Privilege escalation from user to admin is called?

- A. Horizontal
- B. Vertical
- C. Passive
- D. Internal

23. Event ID 4625 represents?

- A. Successful login
- B. Failed login
- C. Account deletion
- D. Password reset

24. Fake Wi-Fi network attack is called?

- A. Evil Twin
- B. Pharming
- C. Smishing
- D. Tailgating

25. Which SOC process analyzes alerts to determine threats?

- A. Alert triage
- B. Encryption
- C. Backup
- D. Patch management

Section B — Short Answer Questions

1. Explain the difference between hashing and encryption.

2. What is password spraying and why is it difficult to detect?

- 3. What is lateral movement and why do attackers perform it?*
- 4. Explain vertical vs horizontal privilege escalation.*
- 5. What is credential stuffing?*
- 6. Explain the concept of salting in password storage.*
- 7. What are Indicators of Compromise (IOC)? Give examples.*
- 8. Explain the difference between phishing and spear phishing.*
- 9. What is an Evil Twin attack?*
- 10. What is alert triage in SOC operations?*

Section C — Scenario Based Questions

- 1. 500 failed login attempts from one IP targeting a single account.*
- 2. One password attempted across 200 accounts.*
- 3. Mimikatz executed → LSASS accessed → Admin login → RDP to file server.*
- 4. Email: 'Your account will be suspended. Click here to reset password.'*
- 5. Login from India followed by Russia within 3 minutes.*

Section D — Analyst Thinking

- 1. What steps would you take investigating a suspicious login alert?*
- 2. If a user clicked a phishing email what should SOC do first?*
- 3. Why do attackers escalate privileges after initial access?*
- 4. How do you differentiate between false positive and real incident?*
- 5. What logs should be checked during a brute force investigation?*

Instructor Answer Key & Explanations

1. What is the primary role of a SOC L1 analyst?

Correct Answer: B

Explanation: SOC L1 analysts monitor alerts, validate them and escalate incidents.

2. Which malware encrypts files and demands payment?

Correct Answer: C

Explanation: Ransomware encrypts files and demands payment.

3. Which attack uses one password across many accounts?

Correct Answer: C

Explanation: Password spraying uses one common password across many users.

4. What does hashing primarily provide?

Correct Answer: B

Explanation: Hashing ensures integrity by verifying data has not changed.

5. What does salting prevent?

Correct Answer: B

Explanation: Salting prevents attackers from using rainbow tables.

6. Which tool is commonly used for credential dumping?

Correct Answer: C

Explanation: Mimikatz extracts credentials from memory.

7. Which Windows process stores authentication credentials?

Correct Answer: C

Explanation: LSASS stores credential information.

8. Which attack redirects users to a fake website without clicking a malicious link?

Correct Answer: B

Explanation: Pharming manipulates DNS to redirect users.

9. Moving between compromised systems inside a network is called?

Correct Answer: C

Explanation: Lateral movement means pivoting between internal systems.

10. Which encryption type uses two keys?

Correct Answer: B

Explanation: Asymmetric encryption uses public/private keys.

11. Fraudulent emails to steal credentials represent which attack?

Correct Answer: A

Explanation: Phishing emails trick users into revealing credentials.

12. What does IOC stand for?

Correct Answer: A

Explanation: IOC indicates evidence of malicious activity.

13. Which attack uses leaked credentials from previous breaches?

Correct Answer: B

Explanation: Credential stuffing uses breached credential lists.

14. Which attack floods servers with traffic?

Correct Answer: C

Explanation: DDoS overwhelms systems with traffic.

15. Which technique extracts credentials from system memory?

Correct Answer: A

Explanation: Credential dumping steals credentials stored in memory.

16. Targeted phishing aimed at a specific individual is called?

Correct Answer: B

Explanation: Spear phishing targets specific victims.

17. What does SIEM primarily do?

Correct Answer: B

Explanation: SIEM collects and correlates security logs.

18. Which security principle ensures data is unchanged?

Correct Answer: B

Explanation: Integrity ensures data has not been altered.

19. Which protocol is used for remote Windows access?

Correct Answer: B

Explanation: RDP enables remote desktop access.

20. Which authentication uses multiple verification factors?

Correct Answer: B

Explanation: Multifactor authentication requires multiple factors.

21. What is alert triage?

Correct Answer: B

Explanation: Alert triage evaluates alerts to determine severity.

22. Privilege escalation from user to admin is called?

Correct Answer: B

Explanation: Vertical escalation increases privilege level.

23. Event ID 4625 represents?

Correct Answer: B

Explanation: 4625 indicates failed login attempts.

24. Fake Wi-Fi network attack is called?

Correct Answer: A

Explanation: Evil Twin mimics legitimate Wi-Fi networks.

25. Which SOC process analyzes alerts to determine threats?

Correct Answer: A

Explanation: Alert triage is the first SOC investigation step.